

MAHESH PANTA

Nepal

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SUMMARY

Computer Engineering undergraduate at Kathmandu University with hands-on experience in machine learning, computer vision, and 3D reconstruction research. Currently a Research Trainee at KU's HI Lab, working on depth estimation and 3D scene reconstruction. Proficient in building deep learning pipelines (TensorFlow, PyTorch), training object detection models (YOLOv8), and deploying ML-powered APIs with FastAPI and PostgreSQL. Passionate about applying computer vision, reinforcement learning, and full-stack AI to real-world problems.

EDUCATION

Kathmandu University

Bachelor of Engineering in Computer Engineering

Dhulikhel, Nepal

Present — Completed 2 Semesters

Khwopa Secondary School

NEB Plus Two (Science with Biology) — GPA: 3.7

Graduated

EXPERIENCE

Research Trainee — Computer Vision & 3D Reconstruction

Mar 2026 – Present

HI Lab, Kathmandu University

Dhulikhel, Nepal

- Conducting research in computer vision and 3D scene reconstruction as part of KU's Human-Intelligence Lab.
- Working on depth estimation, point cloud generation, and 3D structure-from-motion (SfM) pipelines using Python and OpenCV.
- Exploring neural radiance fields (NeRF) and multi-view geometry techniques for photorealistic 3D reconstruction.
- Collaborating with lab members on applied CV research targeting real-world spatial understanding problems.

TECHNICAL SKILLS

Programming Languages: Python, C, C++

Machine Learning & AI: Supervised & Unsupervised Learning, CNNs, Transfer Learning, Data Augmentation, Model Optimization, Recommender Systems

Computer Vision: YOLOv8 (Object Detection & Segmentation), OpenCV, 3D Reconstruction, Depth Estimation, Structure-from-Motion, NeRF

Reinforcement Learning: Q-Learning, Model-Based Agents, Reflex Agents, Environment Simulation

Deep Learning Frameworks: TensorFlow, Keras, PyTorch (fundamentals)

Data Science: NumPy, Pandas, Matplotlib, Scikit-learn

Backend & APIs: FastAPI, Flask, REST API Design

Databases: PostgreSQL, SQLite, SQLAlchemy

Mobile & Frontend: Flutter (Dart), Kivy, HTML, JavaScript (Beginner)

Game Development: C++ with Raylib, RayGUI

Networking: ENet (Reliable UDP), REST APIs

Tools & OS: Git, CMake, Ubuntu, Jupyter Notebook

Languages: English, Nepali, Hindi

PROJECTS

ChitraDartaa — Civic Issue Reporting System | *Flutter, FastAPI, PostgreSQL, YOLOv8, TensorFlow* 🐙 GitHub

- Built a full-stack civic issue reporting platform enabling citizens to submit geo-tagged image reports (potholes, garbage, road damage) with GPS location and resolution status tracking.
- Trained **YOLOv8** object detection and segmentation models to automatically classify issue types from uploaded images.
- Developed CNN-based classifiers (road-issue vs non-issue, potholes vs garbage) with TensorFlow/Keras, applying data preprocessing and augmentation pipelines.
- Designed a **FastAPI** REST backend with **PostgreSQL** (via SQLAlchemy) for persistent storage of reports, user data, and resolution tracking.

- Built a cross-platform **Flutter** mobile frontend with interactive map integration, push notifications, and an admin dashboard for authorities.

Forest Conservation System | *Python, Computer Vision, Machine Learning*

 GitHub

- Developed an AI-powered forest monitoring system for detecting deforestation and environmental threats from image and remote sensing data.
- Applied computer vision techniques for change detection and classification of forest cover using deep learning models.
- Integrated ML pipelines for automated environmental risk assessment.

GameVault — Mini Game Arcade with AI Agents | *C++, Raylib, ENet*

 GitHub

- Co-developed a multi-game arcade (Pong, Snake, Space Invaders) with real-time multiplayer via ENet reliable UDP networking.
- Implemented RL-inspired AI agents: a Simple Reflex Agent for reactive play and a **Model-Based Agent** for stateful environment tracking.
- Managed cross-platform builds and dependency resolution using CMake.

YummY-Chef Chatbot | *Python, Kivy, ChatterBot, SQLite*

 GitHub

- Built a mobile-friendly chatbot providing recipe suggestions and cooking assistance using ChatterBot NLP and a Kivy cross-platform UI.
- Implemented persistent data storage and conversation memory with SQLite.

CERTIFICATIONS

Unsupervised Learning, Recommenders, Reinforcement Learning

Apr 2026

DeepLearning.AI — Andrew Ng | Credential: 49W64MVQ5LQ9

Machine Learning Specialization

Apr 2026

DeepLearning.AI — Andrew Ng | Credential: JUBLWYJD7HKT

Advanced Learning Algorithms

Nov 2025

DeepLearning.AI — Andrew Ng | Credential: BQCH8WWP8VMT

Supervised Machine Learning: Regression and Classification

Oct 2025

DeepLearning.AI — Andrew Ng | Credential: U0574LHEIT2T

Introduction to Programming Using Python

CS50, Harvard University

Pandas

Kaggle